

# JESSE ANGELINA PALAKONDA

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## PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science Engineering student at VIT Amaravati (CGPA: 8.66) with hands-on programming experience in Python and SQL, and a strong record of building, testing, and shipping complete software projects end to end. Comfortable learning new languages and tools quickly, working independently and in teams, and navigating ambiguous requirements to deliver working solutions. Author of 3 IEEE publications and AWS Certified Cloud Practitioner. Seeking to grow as a software engineer through Google's Software Application Development Apprenticeship.

## TECHNICAL SKILLS

- **Programming:** Python, Java, SQL (queries, joins, subqueries, aggregations, optimization), basics of Flask
- **Software Practices:** unit testing, data validation, debugging, version control (Git/GitHub), coding standards & documentation
- **AI/ML & Data:** Machine Learning, Deep Learning, TensorFlow, Keras, PyTorch, YOLOv8, OpenCV, pandas, NumPy
- **Tools & Productivity:** Google Sheets, MS Excel, Power BI (basics), Jupyter Notebook, VS Code
- **Cloud:** AWS (Certified Cloud Practitioner)

## PROJECTS

- Gesture & Voice Controlled Virtual Mouse** | *Python, OpenCV, MediaPipe, SpeechRecognition* 2025
- Built a real-time desktop application translating hand gestures and voice commands into mouse actions, combining computer vision and speech-recognition modules into one working system.
  - Wrote and ran iterative test cycles to tune recognition thresholds, reaching 92.8–94.5% gesture accuracy and up to 96.7% voice-command accuracy.
- Multimodal Kidney Disease Detection & Segmentation** | *Python, PyTorch, TensorFlow, U-Net++, OpenCV* 2025
- Designed and implemented a multimodal deep-learning pipeline processing CT and MRI data, with dedicated preprocessing and validation stages to keep inputs consistent across modalities.
  - Implemented U-Net-based architectures and feature-fusion logic; verified reliability through structured evaluation, reaching a combined mean Dice score of 0.9022.
- Brain Tumor Detection & Segmentation** | *Python, YOLOv8, OpenCV* 2024
- Built an end-to-end detection pipeline (preprocessing, augmentation, validation) to keep training inputs high-quality and consistent.
  - Achieved 84.2% mAP@50 and 81.7% mask precision; documented results and testing methodology for reproducibility.
- Real-Time Object Detection using YOLOv9c and Flask** | *Python, YOLOv9c, Flask, OpenCV* 2024
- Built a Flask web application serving a YOLOv9c object-detection model across three input modalities: static images, uploaded video, and live webcam streams.
  - Validated model performance across modalities, achieving 94.7% accuracy on images, 93.5% on video, and 92.8% on webcam detection.

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Vellore Institute of Technology (VIT), Amaravati - B.Tech, Computer Science Engineering</b><br>CGPA: 8.66 / 10 | 2022 - 2026 |
| <b>Torque Junior College - Intermediate (MPC)</b><br>96.3%                                                        | 2020 - 2022 |
| <b>Timpany Steel City School - Class X (CBSE)</b><br>89.2%                                                        | 2019 - 2020 |

## RESEARCH PUBLICATIONS (IEEE)

- Gesture and Voice-Controlled Virtual Mouse - IEEE INDIACOM 2025
- Real-Time Object Detection Using YOLOv9c and Flask - IEEE DELCON 2024
- Object Detection using Synthetic Data and Generative AI - IEEE ICSCDS 2025

## CERTIFICATIONS

- AWS Certified Cloud Practitioner - Amazon Web Services, 2025
- Google Data Analytics Professional Certificate - Coursera, 2024
- IBM Data Engineering Professional Certificate - Coursera, 2025